Implications of Early Devonian poles from the Canadian Arctic Archipelago for the North American apparent polar wander path[1]

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Lower Devonian red beds from the upper member of the Peel Sound Formation at Prince of Wales Island (Canadian arctic) yield two different paleopoles at 25N 099E and at 01N 091E, the first one being older than the second one. The magnetic directions from which the poles are calculated are derived from vector analysis of thermal, chemical, and alternating magnetic field demagnetization results. Normal and reversed polarities are recorded for the northerly pole, whereas the pole at the equator reveals mainly a normal polarity. The position of the pole close to the equator has significant implications for the early Paleozoic apparent polar wander path of the North American craton. It appears that from the Late Cambrian to Early Devonian the craton moved continuously in a counter-clockwise direction that ended abruptly in the Early Devonian when the direction of the motion of the continent was reversed in a very similar manner to what occurred in Late Cambrian times.

Les couches rouges d'âge Dévonian inférieur appartenant au membre supérieur de la formation de Peel Sound à l'île Prince of Wales (arctique canadien) ont fourni deux paléopôles différents à 25N 099E et à 01N 091E, le premier étant antérieur au second. Les directions magnétiques ayant servi à calculer les pôles découlent d'une analyse des vecteurs obtenus à partir des résultats de la désaimantation par des traitements thermiques, chimiques et par des champs alternatifs. Des polarités normales et inversées ont été enregistrées pour le pôle orienté vers le nord, tandis que le pôle à l'équateur indique principalement une polarité normale. La position du pôle près de l'équateur est importante pour déterminer, au Paléozoïque inférieur, le tracé de la dérive apparente des pôles du craton de l'Amérique du Nord. Il semble que, du Cambrien supérieur jusqu'au Dévonien inférieur, le craton ait glissé continuellement dans la direction inverse du sens des aiguilles d'une montre, que ce mouvement ait pris fin brusquement au Dévonien inférieur lorsque la direction de déplacement s'est inversée de manière analogue à ce qui se produisit aux temps du Cambrien supérieur.

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Introduction

The number of paleopoles of Late Cambrian to Early Devonian age of the stable North American craton is very limited. For a time span of about 100 Ma only seven paleopoles are considered to be reliable by Van der Voo *et al.* (1980), which is too few to represent the complete apparent polar wander (APW) path for that period. If poles of lower Paleozoic rock units from the eastern North American seaboard are also included, many more data are available (Roy *et al.* 1979). However, the scatter of data is very large (up to 70°). At this stage neither for the craton nor for the seaboard can an APW path be calculated that is accurate enough to establish relative motions between the two areas during the early Paleozoic. Among the processes that might have caused the large scatter of the poles are: relative motions of blocks (Kent and Opdyke 1978), rapid polar wander (Roy *et al.* 1979), remagnetization during the Carboniferous (Roy *et al.* 1979), or a combination of all the above. In trying to find a solution for the early Paleozoic APW path it is firstly important to obtain reliable data from areas that belong without any doubt to the North American craton and that have not been remagnetized during their geological history. Rocks that most likely meet these criteria are red sediments of the Peel Sound Formation (Thorsteinsson and Tozer 1963) of Siluro-Devonian age, which are found in the Canadian Arctic Archipelago at the northern margin of the North American craton. These sediments are practically undisturbed tectonically and no other geological event of any kind that might have affected the magnetic properties has occurred in that area since deposition of the sediments.

Geology

Along the northern margin of the North American craton sediments of Cambrian age and younger are deposited upon Precambrian basement rocks. The latest tectonic activity affecting basement rocks ended in the Middle Silurian (Thorsteinsson 1980) and resulted in a north–south trending ridge between the Boothia Peninsula and Cornwallis Island called the Boothia uplift. On the flanks of this topographic high, sediments ranging in composition from conglomerates to siltstones occur that gradually change laterally into the dolomites and carbonates of the miogeosynclinal structure of the Inuitian Orogeny to the north. Detritus from the Boothia uplift

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formed a sequence of Siluro-Devonian sediments belonging to the Peel Sound Formation (Thorsteinsson and Tozer 1963). These sediments are exposed at eastern Prince of Wales Island. In the Peel Sound Formation two members can be distinguished (Thorsteinsson and Tozer 1963). The lower member consists of conglomerates and is unconformably overlain by the upper member made up of conglomerates, sandstones, and siltstones. At Prince of Wales Island it is predominantly the upper member that is exposed. Its strata are nearly horizontal with a slight, mostly westerly dip of only a few degrees. This dip is regarded as being a primary one because syndepositional vertical movements of the Boothia uplift created stronger relief and deposition under high energy conditions on a steep depositional slope (Miall and Gibling 1978). The upper member of the Peel Sound Formation has been interpreted as deltaic alluvial fan and piedmont deposits with paleocurrent directions towards the west. Miall (1970) distinguished several facies belts, which are, from east to west: (1) conglomerate facies; (2) conglomerate–sandstone facies: interbedded conglomerate, fine to coarse red sandstone and red siltstone, deposited by low sinuosity braided streams; (3) sandstone facies: point bar deposition in high sinuosity streams; (4) sandstone–carbonate facies: red, pink, and buff sandstones, interbedded with dolomites deposited in an estuarine environment. Miall (1970) gave a fifth facies, the carbonate facies, but this is now considered to belong to the Drake Bay Formation (Thorsteinsson 1980). Paleomagnetic samples were collected from facies zones 2, 3, and 4.

Age

The upper member of the Peel Sound Formation is of Early Devonian (late Lochkovian) age (Thorsteinsson 1980). Evidence comes from fossils collected in the sandstone–carbonate facies near the north coast of Prince of Wales Island. The fossils are mostly vertebrates (a cephalaspid, a large cyathaspid, a small pteraspid, and *Baringaspis dineleyi*) and a conodont (*Pelekysgnathus serratus serratus* Jentzsch) of Early Devonian age and probably not younger than Lochkovian (Thorsteinsson 1980). The upper member of the Peel Sound Formation is also exposed in the northwest of Somerset Island and it can be correlated in time and in lithology with the Snowblind Bay Formation on Cornwallis Island.

Paleomagnetic sampling

At six localities (A–F) at the north coast of Prince of Wales Island paleomagnetic samples were collected (Fig. 1; Table 1). The total distance between the localities is about 10 km. Cores, drilled in the field, had to be oriented with a sun compass because the vicinity of the magnetic pole causes large daily variations of the magnetic declination at the sampling localities.

Outcrops, suitable for paleomagnetic sampling, were mostly found in steep slopes of rivers. Wherever it was possible parallel vertical sections were sampled. At the six localities 246 cores were drilled at 31 sites (Table 1). At localities A and B, which are in the conglomerate–sandstone facies, the sandstones have a pink–red colour, are coarse grained, have intercalated platy mud pebbles and mud layers on top. Locality B is at the same stratigraphic level as the top of section A.

Localities C and D are in the sandstone facies. Coarse- and medium-grained red sandstones alternate with brick-red medium- to fine-grained laminated sandstones. Locality C is equivalent to the top of D. Locality E is topographically higher than C and D and, because of the nearly horizontal position of the beds, stratigraphically younger. The sandstones at locality E are red, fine grained, and laminated. Beds of brown–red siltstones are frequently present but are not easily drilled. The most westerly locality (F) is in the sandstone–carbonate facies. Thick beds of pinkish cross-bedded sandstones alternate with thin dark brown–red siltstone beds. Although locality F is at a topographically lower level than C, D, and E, the time-stratigraphic relationship between the two areas is difficult to assess because of the difference in facies.

Measuring techniques

Measurements of the magnetic remanence were carried out with a JR-4 spinner magnetometer (Jélinek 1966), which is capable of measuring magnetic intensities of 10^{-4} A/m for standard size specimens of 25 mm diameter and 22 mm height. To demagnetize the specimens three different techniques were used: thermal demagnetization, alternating magnetic field (AF) demagnetization, and chemical demagnetization. The demagnetization results were analyzed by using Zijderveld (1967) diagrams. Thermal demagnetization was done stepwise in a large capacity non-inductively wound electric furnace, having a pseudo-zero field of 2 nT (Roy *et al.* 1969, 1972). For AF demagnetization a three-axis tumbler was used (Roy *et al.* 1973). The maximum field that could be reached was 300 mT.

For chemical demagnetization a hole parallel to the cylinder axis was drilled through each specimen to increase the surface exposed to hydrochloric acid (HCl). Leaching of the specimens was done under a vacuum of 10kPa to stimulate penetration of the acid. At several stages the progress of the leaching process was checked by cutting a specimen in half. The strength of the acid was 6 N initially but it was later changed to 10 N (at 3500 h for the first run and at 1500 h for the second run).

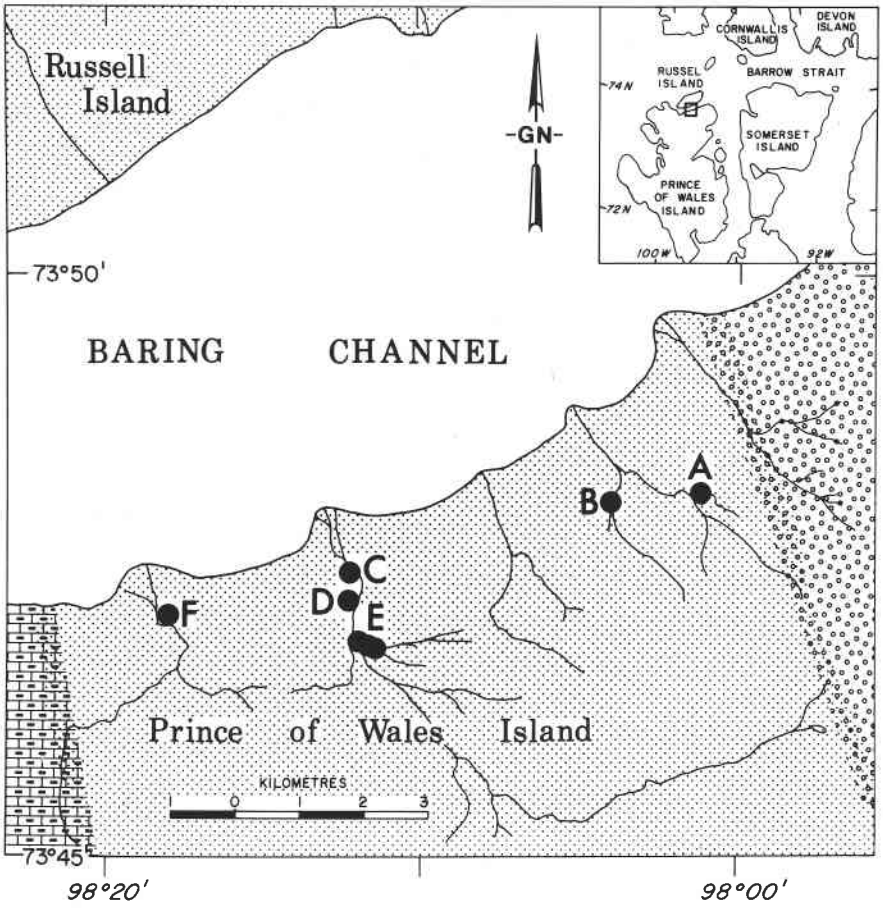


FIG. 1. Sampling localities and facies zones in the Peel Sound Formation on the north shore of Prince of Wales Island.

Paleomagnetic results

The intensity range of the natural remanent magnetization (NRM) is 0.3–6 × 10^{-3} A/m. There is no correlation between the value of the NRM intensity and the type of the rock. All sites reveal scattered NRM directions.

Initial thermal and chemical demagnetizations were done for most of the sites and initial AF demagnetization was done for only a few sites. It appeared that results of all demagnetizations for specimens from localities A and B were either non-interpretable or inconclusive. Zijderveld (1967) diagrams for most of these specimens did not show straight segments or stable end points so that no meaningful direction could be determined. For those specimens from which a direction could be obtained the agreement among the directions was very poor. An explanation might be that the sediments at localities A and B are very coarse and were deposited in a high energy environment, so that the orientation of the magnetic minerals and thus the depositional detrital remanent magnetization (dDRM) are not solely controlled by the Earth's magnetic field but also by the current direction. Taking all this into account no further effort was made to demagnetize specimens from localities A and B.

Better results were obtained from the four other localities. The best separation of multicomponent magnetization was achieved by thermal and chemical demagnetization techniques. For most of the specimens AF demagnetization proved to be inadequate to demagnetize the specimens sufficiently or to get complete separation of components probably because the coercivity spectra of the components are largely overlapping. Therefore, this technique was only used for a minor number of specimens. Initially, a large number of pilot specimens had to be selected from each location for both chemical and thermal demagnetizations because of the wide variety of the rock types. Subsequently the remaining specimens were demagnetized using the technique that had proven to give the most reliable results for a specific rock type at a certain location.

Both single-component and multicomponent magne-

TABLE 1. Sampling and treatments of samples obtained from the localities shown in Fig. 1

Locality	No. of sites	No. of cores	Number of selected specimens demagnetized,[a] and number of reliable directions[b] ( )		
			Thermal	Chemical	AF
A	10	77	10 (—)	11 (—)	3 (—)
B	5	38	4 (—)	5 (—)	1 (—)
C	3	24	6 (6)	15 (13)	3 (2)
D	5	30	10 (10)	21 (16)	7 (3)
E	4	41	18 (23)	12 (8)	6 (1)
F	4	36	7 (8)	15 (13)	6 (2)
	31	246	55 (47)	79 (50)	26 (8)

[a]More than one specimen might have been cut from a core. Only specimens with an NRM well above the detection limit of the JR-4 spinner magnetometer have been used for demagnetization experiments. For these reasons the total number of selected specimens demagnetized can be either larger or smaller than the total number of cores for a specific locality.

[b]The number of reliable directions gives the number of directions that could be derived from demagnetization diagrams using the criteria mentioned in the text. No reliable directions could be derived from a number of specimens, whereas other specimens exhibited more than one clearly defined direction (see Fig. 2). The number of reliable directions can therefore be either smaller or larger than the number of specimens.

tizations are present at most sites. Two different single-component magnetizations could be determined: one northwesterly downward (Nd) and another southeasterly upward (Su). Unblocking temperature curves indicate that both magnetizations are carried by magnetite because the maximum unblocking temperatures are 575°C. When multicomponent magnetizations occur they usually are composed of a magnetite component and a hematite component. In a number of cases these two magnetizations are nearly opposite to each other (Fig. 2*a*). The Su component is then carried by magnetite (unblocking temperatures between 300 and 575°C) and the Nd component by hematite (unblocking temperatures between 575 and 690°C). This type of multicomponent magnetization was mostly found in sites from locality D (Fig. 3). The two components are considered to reflect the same paleofield direction in normal (Nd) and reversed (Su) orientations.

The antiparallelism of the two components is not perfect in all cases (Fig. 2*b*). This is not due to incomplete separation of two oppositely oriented components but is attributable to the presence of another component whose direction is northwest and upward (Nu). That the Nd and the Nu components are two separate magnetizations is revealed by demagnetization diagrams where the Nu magnetization alone is present (Fig. 2*c*) and by other diagrams where both the Nu and the Nd directions are present in the same specimens (Fig. 2*d*). The latter example is a chemical demagnetization. For thermal as well as chemical demagnetizations the Nu component is the hardest. Thermal demagnetization diagrams indicate that the Su magnetization is usually carried by magnetite and the Nd and Nu magnetizations are carried by hematite (Figs. 2, 3). A possible interpretation of this observation is that the Su component represents a (postdepositional) detrital remanent magnetization acquired after deposition but before solidification of the sediment by alignment through rotation of the still mobile magnetic moments of magnetite grains along the Earth's magnetic field direction (Irving and Major 1964). The Nd and Nu magnetizations are probably chemical remanent magnetizations (CRM) acquired after deposition of the sediment when hematite grains grow through their blocking size. The Nd component occurs in the coarse- to middle-grained sandstones such as those from localities C and D. The fact that the Nd and Su directions are nearly anti-parallel indicates that a field reversal occurred during the time of acquisition of the two magnetizations. Most of the DRM has been acquired before the reversal and most of the CRM after it. The Nu magnetization is probably also a CRM. Its direction is clearly different from the Nd–Su direction. It is believed that the Nu magnetization has been acquired much later than the Nd–Su direction after a change in field direction relative to the craton. A reason that the Nu magnetization was acquired so much later than the Nd magnetization might be that because of the finer grain size and different structure of the sediment at locality E or because of lesser availability of iron the growth rate of the hematite was slower so that the blocking size was reached later. It means that Su is the oldest magnetization, followed by Nd, and finally Nu. From the stereonets in which every well defined direction is plotted per locality, it is seen that the Nd magnetization is present at localities D and C and the Nu magnetization at E and D. At each locality, at least one south and down direction is recorded. This direction might be the reverse of the Nu direction. However, the number and accuracy are not high enough to give a statistically reliable mean direction. Therefore, these directions have not been used for mean pole calculations. Results from all localities show a smearing from the Su, Nd, and Nu directions towards the east. The difference between the directions is probably too large to be explained by secular variation only. In finding an explanation for the smearing it should be realized that in the whole area transportation directions were from the east to the west. Magnetic minerals that are not equidimensional in shape might therefore have been deposited in a preferred orientation, thus causing a

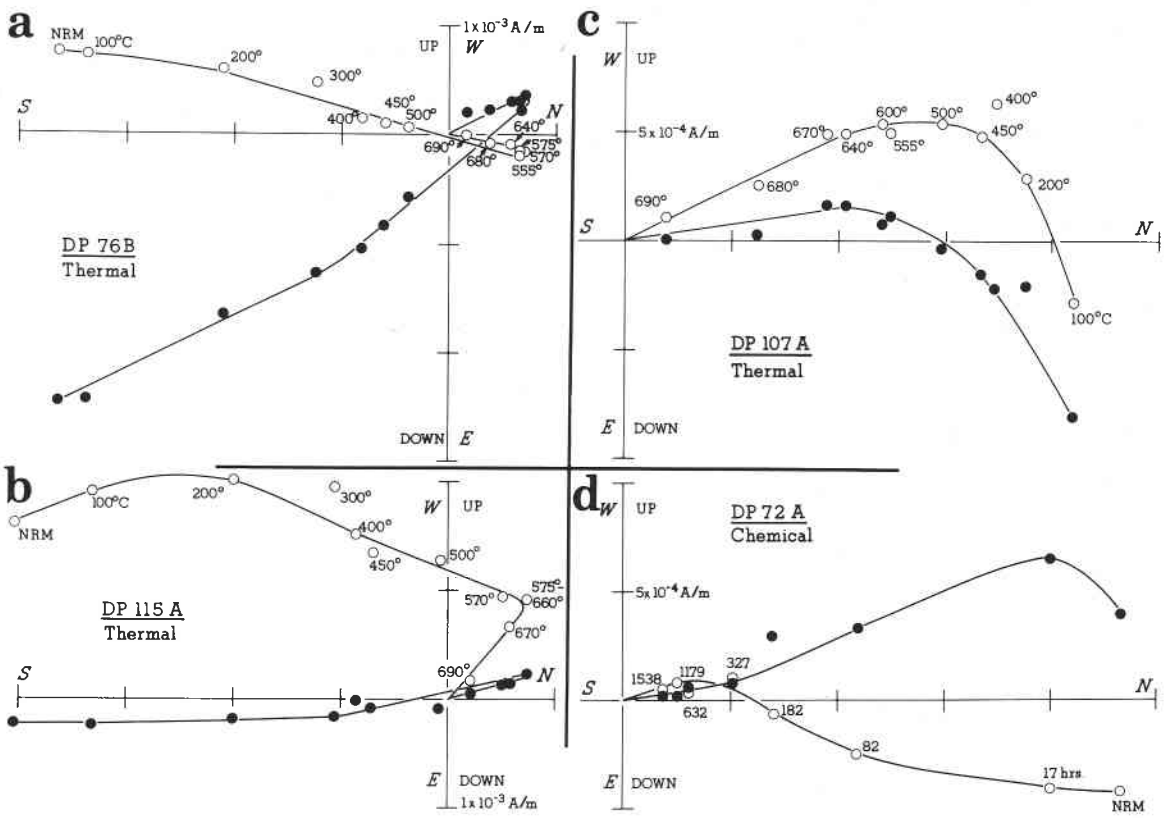


FIG. 2. Thermal and chemical demagnetization diagrams (Zijderveld 1967). Solid symbols are projections on the horizontal plane, open symbols on the vertical north–south plane. All diagrams show the multicomponent nature of the magnetization. The magnetite magnetization (below 575°C) could easily be separated from the hematite magnetization (between 600 and 690°C). For clarity the negative inclination in sample DP 72A (*d*) has been slightly exaggerated. The real direction of the hard magnetization is 325, −12.

biased DRM direction. Because of this uncertainty, the poles calculated from these easterly directions are not used in the following discussion.

Means per site and per locality were calculated for each of the three directions giving specimens unit weight (Table 2). The slight regional tilt of the beds in the area is most likely a primary tilt of sedimentary strata deposited in a terrestrial facies on the slope of the Boothia uplift. There is no indication that the dip of the strata is caused by tectonic movements. Therefore no correction for this effect should be made. Paleopoles calculated from the magnetic directions (Table 2) are plotted in Fig. 4. The Su poles are close to each other and the Nd pole of locality D fits very well in this group. The Nd pole from locality C is farther to the north, probably biased by the present-day magnetic field, which is nearly vertical. It is therefore expected that the best estimate of the true paleopole for the Early Devonian (M_{DL}) is given by calculating the mean from C (Su), D (Nd and Su), E (Su), and F (Su). The Nu direction of locality E gives a paleopole that does not belong to this group.

Discussion

The pole positions derived from the Peel Sound Formation are not anywhere near paleopoles of younger age. This supports the belief that we are dealing here with poles of Early Devonian age. Their position fits very well with Silurian poles such as those of the Bloomsburg Formation (Roy *et al.* 1967), the Rose Hill Formation (French and Van der Voo 1977), and the Chapman Ridge Formation (Watts and Van der Voo 1979). The trend of the APW path for the North American craton that starts to move southward in the Late Ordovician as proposed by Van der Voo *et al.* (1980) appears now to continue its movement through the Silurian and Early Devonian, following the path passing through M_{DL} to E at the equator. Preliminary paleomagnetic results from the Peel Sound Formation of Somerset Island and Cornwallis Island confirm this trend (P. L. Lapointe, personal communication, 1981). From the equator, the APW curve almost reverses its path and goes to the Middle and Late Devonian poles of the Delaware and Columbus Limestones (Martin 1975),

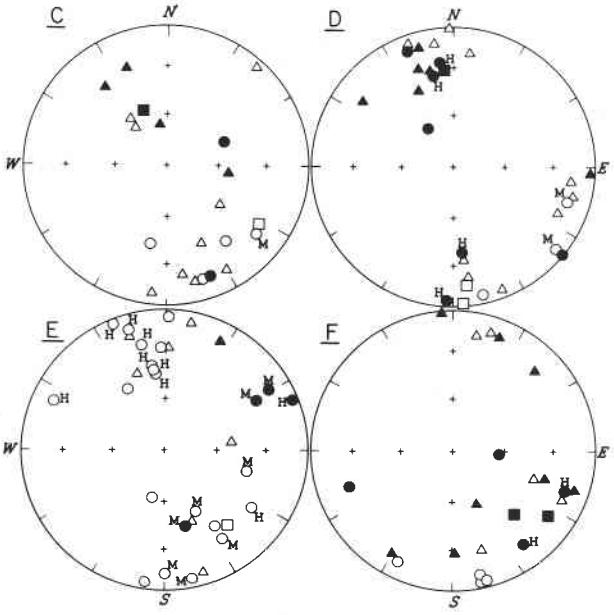


FIG. 3. Equal-area projections of specimen directions from localities C, D, E, and F after thermal (circles), chemical (triangles), and AF (squares) demagnetizations. Solid symbols denote downward inclinations and open symbols upward inclinations. M and H for thermal demagnetization directions indicate that the magnetization is carried by magnetite and hematite, respectively.

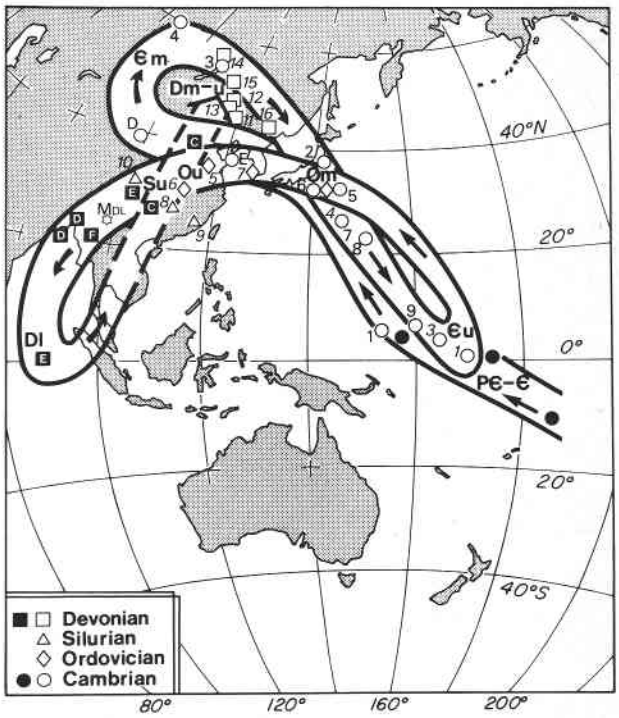


FIG. 4. APW path from the Precambrian–Cambrian boundary to the Late Devonian. The data that are used for reconstructing the path are from Dankers and Lapointe (1981) for the Precambrian–Cambrian boundary (solid symbols), from Watts *et al.* (1980*a*,*b*) for the Cambrian, from Van der Voo *et al.* (1980) for the Late Cambrian to Late Silurian and for the Middle to Late Devonian. Numbers and letters in the diagram are the same as used in these studies. The Early Devonian poles from the present study are indicated by squares.

the Catskill Formation (Kent and Opdyke 1978), the Martin Formation and the Temple Butte Limestone (Elston and Bressler 1977), and the Catskill Formation (Van der Voo *et al.* 1979).

The position of the new poles found in the Lower Devonian sediments from Prince of Wales Island changes the configuration of the early Paleozoic APW curve considerably. The complete early Paleozoic APW path is given in Fig. 4. To avoid complications, only poles of high quality belonging to stable parts of the North American craton are used. These paleopoles and the APW curve derived from them are the same as used by Watts *et al.* (1980*a*,*b*) for the Early Cambrian to the Late Cambrian and by Van der Voo *et al.* (1980) for the Late Cambrian to the Silurian. The continuation of the APW path during the Devonian has to go through the Early Devonian poles of the Peel Sound Formation and might subsequently turn sharply to the northerly poles of the Middle to Late Devonian. At two places in this APW path a sharp kink occurs and the path is almost reversed. Both points are on the equator. The APW path arrives at the equator in Late Cambrian times, turns sharply, and follows a northerly route to arrive again at the equator in the Early Devonian, when it turns again sharply to the north.

The most important features of the APW path presented here are the abrupt change in direction during the Early Devonian and the large polar shift. Neither of these two features is exceptional. The configuration of the APW path in the Devonian is very similar to the one in the Cambrian and the rate of the apparent polar shift in the Devonian is of the same order of magnitude as in the Cambrian.

A number of Devonian poles reported in the literature do not fall on the proposed APW path. For example, poles from the Mount Peyton granite and diorite (Lapointe 1979), the St. George and St. Stephen plutons (Roy *et al.* 1979), and the Bonavista Bay dikes (Murthy and Rao 1979, quoted in Roy *et al.* 1979) all fall in southern latitudes and nowhere near the proposed northern route of the APW path. All the rocks from which these poles are derived have in common is that they are igneous rocks occurring at the eastern North American seaboard. The deviation might originate from either of these two properties, or from both.

TABLE 2. Compilation of mean direction statistics and paleopoles used in Fig. 4

Area		N	$D°$ $I°$	k	α_{95}	Pole position	$d_p°$	$d_m°$
C	Su	12	156 −26	9	15	28N 108E	9	17
	Nd	4	334 +44	16	23	40N 113E	18	29
		5	Not used					
D	Su	7	172 −11	22	13	22N 091E	7	13
	Nd	13	349 +16	14	11	24N 093E	6	12
		9	Not used					
E	Su	11	160 −28	11	14	30N 104E	8	16
	Nu	13	351 −28	18	10	01N 091E	6	11
F		8	Not used					
	Su	3	165 −12	32	22	22N 098E	11	22
		19	Not used					
M_{DL}[a]		5	165 −19	66	9	25N 099E	5	10

NOTES: N is the number of specimens; D and I are declination and inclination, respectively; k is the precision estimate; and α_{95} is the radius of confidence at $P = 0.05$; d_p and d_m are the semi-axes of the oval of confidence at $P = 0.05$.

[a]M_{DL} is the average of C (Su), D (Nd) opposite polarity, D (Su), E (Su), and F (Su).

Acknowledgments

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